ABSTRACT

SIH Problem Statement No.: 26087

Title: *CoopSync AI: AI-Enabled Cooperative ERP, Capacity Building & Employment Ecosystem*

The proposed project focuses on developing **CoopSync AI**, an end-to-end integrated digital enterprise platform tailored for cooperative training institutes, learning management, and employment ecosystems. Cooperative educational entities frequently encounter structural fragmentation—relying on manual attendance registers vulnerable to proxy marking, isolated course tracking, unverified paper certification susceptible to forgery, and disconnected job placement pathways.

The system introduces a **dual biometric and cryptographic verification architecture** for attendance management. It integrates a computer-vision biometric verification pipeline utilizing **Haar Cascades** for face detection, **Local Binary Pattern Histograms (LBPH)** for facial identification, and **Laplacian variance texture analysis** for anti-spoofing and liveness verification. To ensure flexibility across diverse environments, a secondary **dynamic QR code mechanism** provides time-bounded (30-second expiring) session tokens fortified with **SHA-256 cryptographic signatures**, duplicate check-in prevention, and offline synchronization capabilities.

Beyond attendance governance, the platform unifies an academic **Learning Management System (LMS)** with an automated, tamper-proof **digital certification engine**. Certificates are dynamically generated with cryptographically signed verification QR codes for instant public verification. Furthermore, an **AI-driven employment matching engine** algorithmically assesses candidate skill portfolios against cooperative industry requirements to facilitate targeted recruitment, complemented by an interactive **AI career advisory chatbot** and real-time operational analytics dashboards.

The main objective is to demonstrate a secure, scalable, and modular cooperative ERP solution that eradicates proxy attendance, enforces verifiable credentials, and bridges the gap between vocational training and cooperative employment.

Keywords: Computer Vision, LBPH Face Recognition, Anti-Spoofing, Dynamic QR Tokens, Cooperative ERP, Learning Management System (LMS), Tamper-Proof Certification, Skill-Matching Engine, Smart Education.

Figure 1: CoopSync AI Software Architecture & Modular Workflow

CoopSync AI

Software Workflow

Integrated Cooperative Training, Learning & Employment Platform

